Neea Rusch

Curriculum vitae

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Research Interests

Programming languages, formal methods, implicit computational complexity, static program analysis.

Education

8/2021–12/2025 Doctor of Philosophy, Computer & Cyber Sciences, *Augusta University*, Augusta, GA USA.
8/2014–5/2016 Master of Science, Computer Science, *Georgia Institute of Technology*, Atlanta, GA USA.
8/2011–5/2014 Bachelor of Science, Computer Science, *Augusta University*, Augusta, GA USA.

Professional Appointments & Employment

10/2025–present Postdoctoral Researcher, *Department Information Technology, Uppsala University*, Uppsala, Sweden.
8/2025–10/2025 Graduate Research Assistant, *The Graduate School, Augusta University*, Augusta GA, USA.
8/2021–5/2024 Graduate Research Assistant, *The Graduate School, Augusta University*, Augusta GA, USA.
7/2020–1/2021 Senior Consultant, *Sparq (formerly Rural Sourcing) Inc.*, Augusta GA, USA.
1/2019–12/2021 Instructor, *School of Computer & Cyber Sciences, Augusta University*, Augusta GA, USA.
8/2018–5/2019 Android Developer, *TaxSlayer*, Augusta GA, USA.
5/2014–8/2018 Software Developer, *TaxSlayer*, Augusta GA, USA.

Publications

2025 Neea Rusch. “Applied Implicit Computational Complexity”. PhD thesis. Augusta, Georgia, United States: Augusta University, Sept. 2025. DOI: 10.5281/zenodo.17148449.

2023 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. “Distributing and Parallelizing Non-canonical Loops”. In: *Verification, Model Checking, and Abstract Interpretation*. Springer Nature Switzerland, 2023, pp. 1–24. DOI: 10.1007/978-3-031-24950-1_1.

2023 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. “pymwp: A Static Analyzer Determining Polynomial Growth Bounds”. In: *Automated Technology for Verification and Analysis*. Springer International Publishing, 2023, pp. 263–275. DOI: 10.1007/978-3-031-45332-8_14.

2022 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. “mwp-Analysis Improvement and Implementation: Realizing Implicit Computational Complexity”. In: *7th International Conference on Formal Structures for Computation and Deduction (FSCD 2022)*. Vol. 228. LIPIcs. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2022, 26:1–26:23. DOI: 10.4230/LIPIcs.FSCD.2022.26.

2022 Neea Rusch. “Formally Verified Resource Bounds Through Implicit Computational Complexity”. In: *Companion Proceedings of the 2022 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity*. SPLASH Companion 2022. ACM, 2022, pp. 17–20. DOI: 10.1145/3563768.3565545.

Software

2024 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. *pymwp: MWP analysis in Python*. Version 0.5.2. Nov. 2024. DOI: 10.5281/zenodo.7879822.

2022 Clément Aubert, Thomas Rubiano, Neea Rusch, and Thomas Seiller. *Distributing and Parallelizing Non-canonical Loops – Artifact*. Version 1.0. Sept. 2022. DOI: 10.5281/zenodo.7080145.

Research Presentations

Presentation titles and slides are available at `neea.pl/posts/talks`.

2025-07-03 Doctoral Symposium at ECOOP 2025, *Høgskulen på Vestlandet, Bergen, Norway.*
2025-06-03 Programming Languages Seminar, *Uppsala University, Uppsala, Sweden.*
2024-10-14 The 19th Workshop on Programming Languages and Analysis for Security (PLAS'24), *Salt Lake City UT, USA.*
2024-08-20 Theory Seminar, *Aalto University, Espoo, Finland.*
2024-08-07 Lightning talk at the Marktoberdorf Summer School, *Herrsching, Germany.*
2024-07-10 5-min. talks at the Computer Security Foundations Symposium, *University of Twente, Enschede, the Netherlands.*
2024-04-23 The 3-Minute Thesis (3MT), research communication competition, *Augusta University, Augusta GA, USA.*
2023-10-25 The 21st International Symposium on Automated Technology for Verification and Analysis (ATVA'23), *Singapore Management University, Singapore.*
2023-10-20 The International Conference on Security and Privacy in Communication Networks (SecureComm'23), *Hong Kong Polytechnic University, Hong Kong SAR.*
2023-09-15 Lightning talk at the International Autumn School on Proof and Computation, *Herrsching, Germany.*
2023-03-10 SCOT Webinar: A Seminar on Semantic and Formal Approaches to Complexity, *online.*
2023-01-21 The Ninth International Workshop on Coq for Programming Languages (CoqPL'23), *Boston MA, USA.*
2023-01-16 The 24th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI'23), *Boston MA, USA.*
2022-12-06 The 2022 ACM SIGPLAN International Conference on Systems, Programming, Languages, and Applications: Software for Humanity (SPLASH'22) Doctoral Symposium, *University of Auckland, New Zealand.*
2022-06-20 The 28th International Conference on Types for Proofs and Programs (TYPES'22), *University of Nantes, France.*
2022-04-01 The 37th Annual Graduate Research Day – poster, *Augusta University, USA.*
2021-11-18 Laboratoire d'Informatique de Paris Nord (LIPN) seminar, *Université Sorbonne Paris Nord, France.*
2021-11-15 Inria Celtique research team seminar, *IRISA Rennes, France.*
2021-03-26 The 36th Annual Graduate Research Day, *online, Augusta University, USA.*

Professional Services & Reviews

PC Chair **Artifact Evaluation**.
FormaliSE – 2027.

PC Member **Artifact Evaluation Committee**.
APLAS – Asian Symposium on Programming Languages and Systems – 2022.
CAV – Computer Aided Verification – 2024, 2025.
CCS – Computer and Communications Security – 2026.
COORDINATION – International Conference on Coordination Models and Languages – 2023.
ECOOP – European Conference on Object-Oriented Programming – 2023, 2024, 2025.
ICFP – International Conference on Functional Programming – 2022, 2023, 2024, 2025, 2026.
ICSE – International Conference on Software Engineering – 2024, 2027.
iFM – Integrated Formal Methods – 2025.
OOPSLA – Object-oriented Programming, Systems, Languages, and Applications – 2024, 2025, 2026.
PLDI – Programming Language Design and Implementation – 2022, 2023.
POPL – Principles of Programming Languages – 2024, 2025, 2026, 2027.
PPoPP – Principles and Practice of Parallel Programming – 2023.
SOSP – Symposium on Operating Systems Principles – 2023.
TACAS – Tools and Algorithms for the Construction and Analysis of Systems – 2025, 2026.
⟨Programming⟩ – Art, Science, and Engineering of Programming – 2025.

PC Member **Program Committee**.
ICE – Interaction and Concurrency Experiences workshop – 2026.
ECOOP Research Papers, Extended Review Committee – 2023, 2024.
MSR – Mining Software Repositories, Junior Program Committee – 2027.

Editor **Journal Editor**.
JOSS – The Journal of Open Source Software – 2025–present.

Reviewer **Journal Reviews**.
JOSS – The Journal of Open Source Software – 2023–2026.
Sci. Comput. Program. – Science of Computer Programming – 2024.

Organization Committee.
Organizer SERPL – The Southeast Regional Programming Languages Seminar – 2023.
VerifyThis – competition co-organizer – 2027.
Volunteer Student volunteer at POPL'25, ETAPS'24, CAV'23, POPL'23, and SPLASH'22.
Website chair CAV – Computer Aided Verification – 2026.

Teaching Experiences

Details at `neea.pl/posts/teaching`.

Course Instruction

Graduate/Master level.

2026 Software Testing, *instructor*, Uppsala University.

Undergraduate/Bachelor level.

2019–2021 Principles of Computer Programming I (CSCI 1301), *instructor*, Augusta University.
Total enrollment 228, taught 6 times.

2019 Introduction to Computers and Programming (CSCI 1200), *instructor*, Augusta University.
Total enrollment 24, taught 2 times.

Student Project Supervision

2026 Bachelor's Project, Computer Science, *supervisor*, Uppsala University.
2026 Master's Thesis (3x), Computer Science and Data Science, *subject reviewer*, Uppsala University.
2023 Senior Capstone Project, Computer Science, *co-mentor*, Augusta University.

Academic Awards, Fellowships & Scholarships

Details at `neea.pl/awards` ◦ total monetary awards over US$ 30,000.

10/2026 Distinguished Artifact Reviewer of OOPSLA 2026, *by the AEC chairs.*
4/2026 Distinguished Artifact Reviewer of TACAS 2026, *by the AEC chairs and the ETAPS Steering Committee.*
10/2025 Distinguished Artifact Reviewer of OOPSLA 2025, *by the AEC chairs.*
7/2025 Distinguished Artifact Reviewer of ECOOP 2025, *by the AEC chairs.*
5/2025 Distinguished Artifact Reviewer of TACAS 2025, *awarded by the AEC chairs.*
10/2024 Distinguished Artifact Reviewer of OOPSLA 2024, *awarded by the AEC chairs.*
4/2024 3 Minute Thesis, Contest Finalist, *organized by The Graduate School at Augusta University.*
1/2024 Ralph W. Shrader Graduate Diversity Scholarship, *by AFCEA Educational Foundation.*
1/2023 Student Research Program Award, *by Augusta University Provost's Office and Medical College of Georgia.*
4/2022 Excellence in Research in Computer & Cyber Sciences, *by The Graduate School at Augusta University.*
8/2015 Grace Hopper Scholarship, *by College of Computing, Georgia Institute of Technology.*
4/2014 The Outstanding Computer Science Student Award, Class of 2014, *by Augusta University.*

5/2013 Savannah River Scholars Program Scholarship, *by Augusta University and funded by NSF.*

Schools, Seminars & Workshops

9/2026 Challenges in Future Software Stack Development for AI/HPC, Saint-Hippolyte, France, Sep 28–Oct 2, 2026.

5/2026 Making Replications Count Hackathon, Universität Münster, Germany, May 4–6, 2026.

4/2026 VerifyThis 2026, verification competition, Turin, Italy, Apr 11–12, 2026.

5/2025 Programming Language Implementation Summer School, Bertinoro, Italy, May 26–31, 2025.

8/2024 Marktoberdorf Summer School, Herrsching, Germany, Aug 6–17, 2024.

4/2024 VerifyThis 2024, verification competition, Luxembourg City, Luxembourg, Apr 6–7, 2024.

9/2023 International Autumn School on Proof and Computation, Herrsching, Germany, Sep 10–16, 2023.

7/2023 Verification Mentoring Workshop (VMW) at CAV 2023, Paris, France, Jul 18, 2023.

5/2023 Twelfth Summer School on Formal Techniques (SSFT 2023) & the First Formal Methods in the Field Bootcamp, on principles and practice of formal techniques, Atherton CA, USA, May 24–Jun 2, 2023.

6/2022 Programming Languages Mentoring Workshop, PLMW @ PLDI 2022, San Diego CA, USA, Jun 13–14, 2022.

11/2021 Dagstuhl Seminar 21453, on Static Analyses of Program Flows: Types and Certificate for Complexity, Schloss Dagstuhl, Germany, Nov 7–12, 2021.

6/2021 Oregon Programming Languages Summer School (OPLSS 2021), on foundations of programming and security, online, Jun 14–26, 2021.

4/2021 Midlands Graduate School, in the foundations of computing science, online, Apr 12–16, 2021.

University Service

1/2023–5/2025 Organizer of ΔΛΔ's weekly Programming Languages Reading Group, *Augusta University.*

8/2021–5/2025 Member of The Graduate School's Graduate Student Council (TGS-GSC), *Augusta University.*

9/2023–5/2025 Representative in the Graduate Student Government Association (GSGA), *Augusta University.*

5/2023–8/2024 Vice President and co-founder of ΔΛΔ student organization, *Augusta University.*

9/2023–5/2024 GSGA Funding & Conferences Database Ad Hoc Committee, *Augusta University.*

8/2024–5/2025 GSGA Event Planning Committee, *Augusta University.*

Languages

English, Finnish